

## Original Articles

# Efficiency of invertebrate-based bioassessment for evaluating the ecological status of streams along a gradient of flow intermittence

Julie Crabot<sup>a,b,\*</sup>, Sylvain Dolédec<sup>c</sup>, Maxence Forcellini<sup>a</sup>, Thibault Datry<sup>a</sup>

<sup>a</sup> INRAE, UR RiverLy, Centre de Lyon-Villeurbanne, France

<sup>b</sup> Université Clermont Auvergne, CNRS, GEOLAB, F-63000 Clermont-Ferrand, France

<sup>c</sup> Université Claude Bernard Lyon 1, CNRS, ENTPE, UMR 5023 LEHNA, F-69622, Villeurbanne, France

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## ABSTRACT

Intermittent rivers and ephemeral streams (IRES) represent the majority of the global river network, support high biodiversity and provide multiple ecosystem services. However, they are being degraded at alarming rates, and it may be questioned whether current invertebrate-based biomonitoring protocols, which were designed for rivers and streams that flow all year round, are also appropriate for assessing the ecological status of IRES. Here, we test for the first time the efficiency of the French invertebrate-based biomonitoring indicators to assess the ecological status of intermittent rivers downstream six sewage treatment plants along a gradient of flow intermittence (i.e., the percentage of a year without flow at a given site). We found that the effects of drying confounded with that of sewage and in most cases, current biomonitoring metrics did not distinguish between the effect of sewage and those of drying. Except for the taxonomic richness of invertebrates resistant to drying, all metrics decreased linearly with increasing flow intermittence. In addition, the efficiency for detecting sewage effects was lower at intermittent reaches compared to perennial reaches. Last, we identified biological traits responding to flow intermittence, but not to sewage effluents, and we further discuss the implication for traits-based monitoring. If current biomonitoring protocols calibrated for perennial rivers could still be applied to reaches prone to weak flow intermittence, alternatives are urgently needed for integrating intermittent rivers into current biomonitoring national programs.

## 1. Introduction

In the Anthropocene, water management relies on biomonitoring approaches to assess the responses of aquatic ecosystems to the multiple anthropogenic stressors they are facing (Reid et al., 2019). Accordingly, international legislations such as the Water Framework Directive in Europe (European Commission, 2000) or the Clean Water Act in the USA (Environmental Law Institute, 2012) call for biomonitoring to assess ecological quality of rivers and streams. The use of biological signatures to assess human impairment in streams has a long tradition in freshwater ecology (e.g. reviews in Dolédec and Statzner, 2010; Friberg et al., 2011) and benthic macroinvertebrates are one of the most widely used aquatic organisms for achieving such assessments (see e.g., Bonada et al., 2006a, b; Rosenberg and Resh, 1993). To date, current biomonitoring approaches are based almost exclusively on aquatic communities collected from perennial rivers and streams (Reyjol et al., 2014; Seaman et al., 2016; Stubbington et al., 2018) and suggested alternatives for

intermittent and ephemeral rivers remain rare and are not integrated into national efforts (Leigh et al., 2013; Prat et al., 2014; Steward et al., 2018). Intermittent rivers and ephemeral streams, hereafter IRES, are defined as every rivers and streams that cease to flow and/or dry at some points in space and time (Datry et al., 2016). Many authors have questioned the suitability of current biomonitoring approaches in IRES (Mazor et al., 2014; Munné and Prat, 2011; Prat et al., 2014; Reyjol et al., 2014; Steward et al., 2018; Stubbington et al., 2018; Wilding et al., 2018). However, only a few studies tested the applicability of the current macroinvertebrate-based biomonitoring approaches for intermittent rivers. For example, Morais et al. (2004) showed that the impact of a sewage treatment plant was not detectable in an intermittent stream in Portugal. In addition, a review by Stubbington et al. (2018) suggested that current biomonitoring approaches needed to be refined to be efficient for IRES. Recently, a global quantification of the prevalence of non-perennial rivers indicates that IRES comprise the majority of the river channels on Earth and might represent the norm rather than the

\* Corresponding author at: Université Clermont Auvergne, CNRS, GEOLAB, F-63000 Clermont-Ferrand, France.

E-mail address: [julie.crabot@gmx.fr](mailto:julie.crabot@gmx.fr) (J. Crabot).

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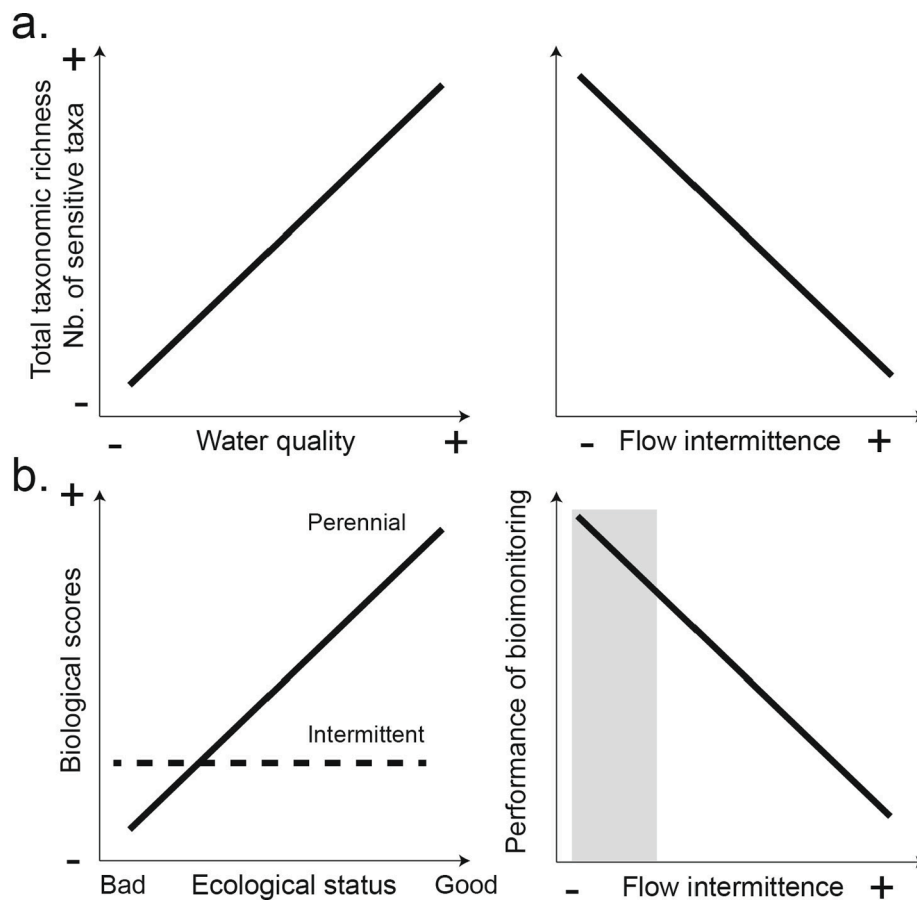
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exception (Messenger et al., 2021). Such a shift in paradigm thus calls for assessing urgently whether current biomonitoring approaches can apply to IRES and if not, the limitations in their use or interpretation can be identified and corrected to accurately reflect the ecological status of the river networks facing the Anthropocene.

Both short- and long-term effects of drying on aquatic communities are relatively well understood due to the exponential number of dedicated studies through the past decade (e.g., Busch et al., 2020; Leigh et al., 2016). Across climatic and biogeographic zones, aquatic alpha-diversity (i.e. the local taxonomic richness at a given location) consistently decreases when annual flow intermittence (i.e. the percentage of a year without flow at a given site) increases (Belmar et al., 2019; Datry et al., 2016; England et al., 2019; Gauthier et al., 2020). Immediately upon drying, taxa without adaptations to cope with desiccation are eliminated locally due to death or migration towards perennial refuges within the landscape (Bogan et al., 2015; Datry et al., 2016). The longer the dry event, the smaller the remaining pool of resistant taxa on the surface (Bogan et al., 2013; Datry, 2012; Pařil et al., 2019) and sub-surface (Storey and Quinn, 2013; Stubbington et al., 2016; Stubbington and Datry, 2013) sediments. Upon rewetting, the arrival of species with high dispersal abilities from refuges also promotes community recovery (Bogan et al., 2015; Datry et al., 2016). With increased drying duration, the distance to refuges tends to increase, thus limiting the recolonization of previously dry sites (Cañedo-Argüelles et al., 2015; Crabot et al., 2020; Gauthier et al., 2020; Sarremejane et al., 2017b). As a result, legacy effects of drying occur even a long time after rewetting and local taxonomic diversity is generally much lower than that of adjacent perennial reaches (Crabot et al., 2021b; Datry, 2012; Gauthier et al., 2020; Soria et al., 2017). Previous studies reported that such taxa loss at

intermittent sites was associated with a lower functional diversity compared to perennial sites (B-Béres et al., 2019; Schriever et al., 2015). However, because biological trait combinations can be expressed by many taxa or by only a few taxa in a local community, functional diversity response to flow intermittence might sometimes differ from the taxonomic patterns. As such, flow intermittence might reduce taxonomic diversity between intermittent and perennial reaches more importantly than functional diversity (Crabot et al., 2021a; Vorste et al., 2016).

The effects of water quality degradation may confound with the natural responses of macroinvertebrate communities to drying (Soria et al., 2020). A large majority of the taxa used as indicators, i.e. with high sensitivity to water quality degradation such as Ephemeroptera, Plecoptera, Trichoptera (EPT) are those that are also primarily affected by drying. For example, the reduction of lateral hydrological connectivity associated with drought leads to a decline in EPT richness (Bonada et al., 2006b). Moreover, the number of EPT taxa differs between permanent and intermittent reference streams (Bogan et al., 2013; Datry, 2012; Straka et al., 2019). In contrast, some taxa known to be resistant to desiccation and widespread in IRES, such as Odonata, Coleoptera and Heteroptera (OCH), but also Oligochaeta and Diptera (Bonada et al., 2006b; Datry, 2012) are more tolerant to different types of water quality degradation, including water pollution (Chang et al., 2014). For example, looking at the 38 indicators taxa used in the French IBG-DCE for their sensitivity to water quality impairment, the only Coleoptera considered are Elmidae and are given a sensitivity of 2 (out of a maximal sensitivity of 9; AFNOR, 2004). Neither Heteroptera nor Odonata are considered in this list. Consequently, we expect a poor success of current biomonitoring approaches in the context of flow intermittence



**Fig. 1.** Conceptualized responses of communities to water quality degradation and flow intermittence (a.) and expected biomonitoring scores and performance in case of perennial and intermittent flow regimes (b.). In grey, the area that could represent an operational space along which biomonitoring approaches developed for perennial rivers could be applicable to intermittent rivers.

(Kalogianni et al., 2017; Soria et al., 2020; Stubbington et al., 2018) since both the number of taxa and taxa showing sensitivity to water quality degradation should decrease with increasing water quality degradation and flow intermittence (Fig. 1a). In turn, the absence of relationships between the measured biological scores and the ecological status may occur in case of flow intermittence, contrary to what is observed in perennial rivers (Munné et al., 2021; Soria et al., 2020) (Fig. 1b). However, as the responses to flow intermittence are known to be linear (e.g. Datry et al. 2014a,b), an effective operational space for weakly intermittent rivers and streams could allow using current biomonitoring approaches (Fig. 1b).

In this study, we quantified the structural and functional community responses to six point-source of sewage along with a gradient of flow intermittence (0 to 60% within a year). We tested the hypothesis that current macroinvertebrate-based biomonitoring protocols were not able to infer the ecological status of naturally intermittent rivers. This is because the effect of drying on communities confounds with those of sewage pollution. Therefore, we first expected that impacted sites (downstream of effluents) located in intermittent reaches would have similar communities to that observed upstream of effluents. Thus, biomonitoring scores would not differ between downstream and upstream sites in the context of flow intermittence, contrary to perennial reaches. Second, we expected the capacity of current macroinvertebrate-based biomonitoring protocols to decrease with increasing flow intermittence, as taxonomic and functional richness are likely to decrease linearly with flow intermittence. Third, we anticipated that some traits should be favored by flow intermittence and some by water quality degradation. To unravel the underlying mechanisms, we identified how biotic and functional traits groups varied with flow intermittence and which one warranted further study over the long-term for establishing robust indicators of intermittent rivers quality.

## 2. Materials and methods

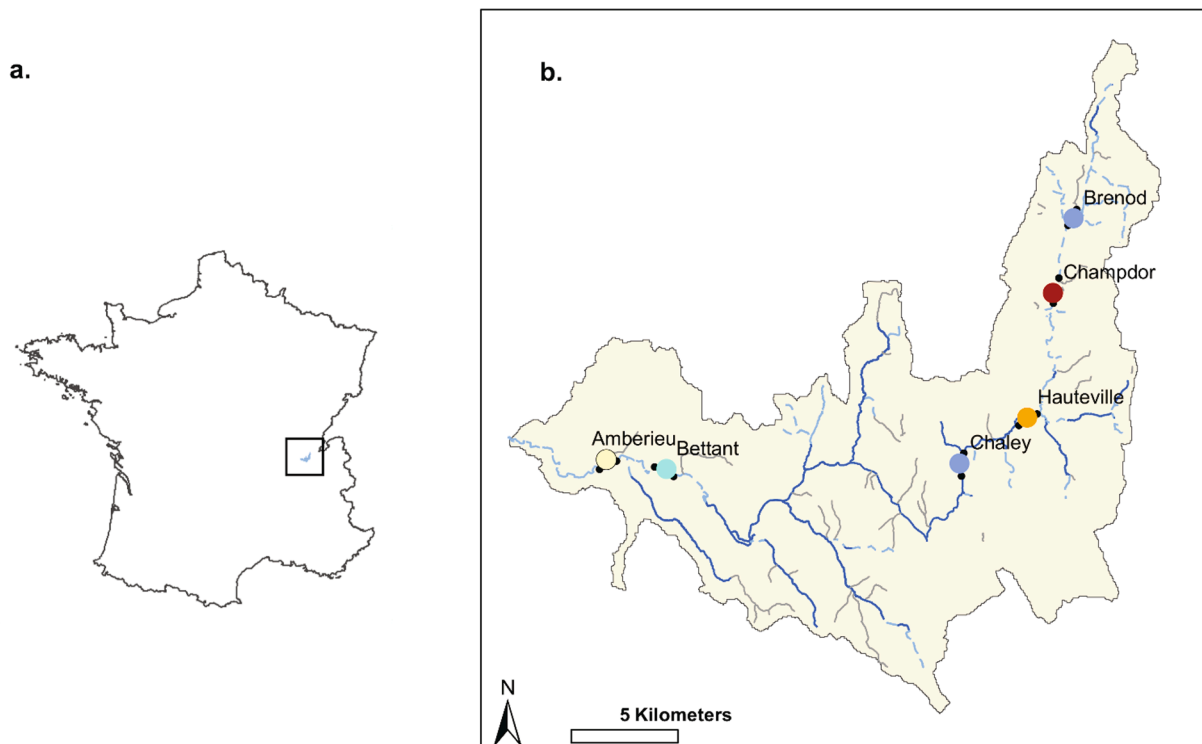
### 2.1. Study sites

The Albarine River is naturally intermittent draining a 313 km<sup>2</sup> catchment located in the temperate eastern France (Fig. 2a). The upper section of the Albarine flows for 45 km on a karstic plateau and through gorges, where several reaches are intermittent due to karstic infiltrations, mostly occurring during the summer periods (Gauthier et al., 2020). Then, the Albarine flows over a 15-km long alluvial plain to finally end in the Ain River, a tributary of the upper Rhône. Surface water continuously infiltrates into the underneath aquifer due to its alluvial plain underlain by thick (>36 m) and highly permeable (from  $2.10^{-2}$  to  $7.10^{-2}$  m s<sup>-1</sup>) sediments and because the river is generally perched 1–14 m above the regional aquifer water table along this section (Datry, 2012; Larned et al., 2011). As a result, this section dries completely during summer, when rainfall cannot compensate for transmission losses into the aquifer (Datry, 2012; Gauthier et al., 2020; Larned et al., 2010). Estimated surface water losses to the aquifer range from 0.85 to 2.5 m<sup>3</sup> s<sup>-1</sup> (Datry, 2012).

In the upper section, depending on the winter snowpack and spring rainfall, drying can occur throughout spring and summer, following a very patchy spatial pattern with high temporal dynamism. Drying events start downstream of the lower sections during spring, at the confluence with the Ain River, and the drying front progressively moves upstream throughout the summer period (Datry, 2012). Flow resumption along the entire river occurs generally at late autumn/early winter (Datry, 2012; Larned et al., 2011). Consequently, annual flow can range between 0 and 60% along the Albarine River (Datry et al., 2011; Larned et al., 2010).

### 2.2. Sampling design

We selected 6 reaches along the Albarine course, each under the



**Fig. 2.** Location of the catchment in France (a), and location of the six sewage treatment plants on the Albarine (b). The color of the circles indicates the flow intermittence at each reach (from dark blue for 0% FI to red for 60% FI) and the small black dots indicate the samplings upstream and downstream of the sewage effluents. Continuous lines represent the perennial sections of the rivers and the dotted lines represent the intermittent sections. Grey section are unknown.

impact of a sewage treatment plant, whose effluents are directly released in the river, and considering the overall gradient of flow intermittence (hereafter FI) ranging from 0 to 60 % (Fig. 2b). At each reach, we selected a site located between 50 and 100 m downstream from the effluent release point and an upstream site located between 50 and 200 m upstream from the effluent release point (Fig. 2b).

### 2.3. Flow intermittence estimates

Flow intermittence was quantified at the 3 reaches located in the upper section using photo-traps (Bushnell Trophycam®) setup for recording one photo every day at midnight and midday and left in place for 2 years (January 2017 until December 2018; Gauthier et al., 2020). For the 3 reaches selected in the lower section, FI was modelled for each reaches using the ELFMOD model (Larned et al., 2011). This model associated flow measurements at rated water-level recorders at each end of the 45-km alluvial section to flow state (flowing or dry) and flow rate (point source discharge measures) at all points between the recorders. The model was calibrated using data from gauging runs between October 2003 and January 2016, and validated using data collected outside of the calibration period. The empirical model was used to reconstruct a 29-yr record of daily flows along the river (Rupp et al., 2008). We subsequently tested the accuracy and precision of model-based estimates of FI using monthly observations of flow made between October 2011 and June 2019 at the selected sampling sites. Modelled FI at each reach was calculated by averaging annual estimates of flow intermittence over the record. Observed FI was highly correlated with modelled FI ( $r = 0.95$ ,  $p < 0.001$ ). The FI took into account only flowing and dry phases because the Albarine River does not present phases of isolated pools, as it is often the case for alluvial rivers with porous streambed sediments (Vorste et al., 2016).

### 2.4. Physical and chemical measurements

The physical and chemical parameters up and downstream effluent releases were sampled at the six reaches during spring (17 April 2018) and fall (19 December 2018) i.e., prior to and after the summer drying period. Water samples and analytical measurements were carried out by the CARSO laboratory (<https://www.groupecarso.com/en/>) that routinely measures the physical and chemical parameters of water bodies within the WFD context. At each location of each reach, 4L of surface water was collected and brought back to the laboratory within 6 h to measure the following physical and chemical parameters according to the national French standards (Rodier et al., 2016): water temperature (°C), pH, specific conductivity ( $\mu\text{S}\cdot\text{cm}^{-1}$ ), dissolved oxygen ( $\text{mg}\cdot\text{L}^{-1}$ ), total phosphorus ( $\text{mg}\cdot\text{L}^{-1}$ , NF EN ISO 9308-3), suspended solid ( $\text{mg}\cdot\text{L}^{-1}$ , NF EN 872), total organic carbon ( $\text{mg}\cdot\text{L}^{-1}$ , NF EN 872), dissolved organic carbon ( $\text{mg}\cdot\text{L}^{-1}$ , NF EN 1484), biochemical demand in oxygen for 5 days ( $\text{mg}\cdot\text{L}^{-1}$ , NF EN 1899-2), chemical oxygen demand ( $\text{mg}\cdot\text{L}^{-1}$ , ISO 15705), Kjeldahl nitrogen ( $\text{mg}\cdot\text{L}^{-1}$ , NF EN 25663), ammonium ( $\text{mg}\cdot\text{L}^{-1}$ , NF T90-015-2), nitrates ( $\text{mg}\cdot\text{L}^{-1}$ , NF EN ISO 10304-1), orthophosphates ( $\text{mg}\cdot\text{L}^{-1}$ , NF EN ISO 68) and nitrites ( $\text{mg}\cdot\text{L}^{-1}$ , NF EN ISO 13395). In addition, organic pollution was measured using the density of *Escherichia coli* (MPN.10  $\text{mL}^{-1}$ , NF EN ISO 9308-3).

### 2.5. Benthic macroinvertebrates

#### 2.5.1. Sampling

Benthic macroinvertebrates were sampled up and downstream effluent releases by the Eurofins Hydrobiology Consulting Company in spring (16-18th April 2018) and fall (17-19th December 2018). Macroinvertebrate sampling was carried out following the NF T90-333 standards (AFNOR, 2016). Briefly, this consisted in the following steps. First, each sampling site was first delineated by estimating the percentage of habitats (e.g. pool, riffles, run), percentage of substrates

(e.g., sand, gravel, cobbles) and percentage of velocity ranges (e.g. slow  $\leq 0.5$  m/s, fast  $> 0.5$  m/s) over a length of approximately 15 times the mean wetted width. This allowed estimating the variety and relative abundance of habitats separated into marginal ( $\leq 5\%$  of the overall sampled location area) and dominant ( $> 5\%$  of the overall sampled location area) substrates. Then, 12 samples were collected up and downstream effluent releases at each reach using a Surber sample ( $1/20 \text{ m}^2$ , 500  $\mu\text{m}$  mesh size) and they were separated into 3 phases: 4 samples into the marginal habitats, 4 samples into the dominant habitats taken in decreasing order of carrying capacity (from the more biogenic – bryophytes, to the last biogenic – artificial uniform surfaces), and 4 complementary samples into the dominant habitats considering their relative proportion over the sampling site. Samples were washed in the field with the coarsest material (sediment, wood) cleaned with a brush and put back in the river. All macroinvertebrates were preserved in 96% ethanol and brought back to the laboratory within the day. Finally, all organisms of each sample were counted and identified to the lowest practical taxonomic level. Most insects and all mollusks were identified to genus or species levels, and crustaceans, annelids and mites to family, order or genus levels following the French standard XP T90-388 (AFNOR, 2010).

#### 2.5.2. Biological indices

Two biotic indexes are currently implemented at the national scale to assess stream impairment in France: the so-called French biotic index (abbreviated IBG-DCE for “Indice Biologique Global Directive Cadre sur l’Eau”; AFNOR, 2016, 2010, 2004) and the French invertebrate multi-metric index (abbreviated I2M2 for “Indice Invertébrés Multi-Métrique”; Mondy et al., 2012). The IBG-DCE enables quantifying the impairment of the water chemistry and the hydromorphological characteristics of a reach by considering the taxonomic richness (TR, grouped into classes ranging from 1 to 14) and the sensitivity of 38 specific indicator taxa (distributed into 9 indicator groups or IG) to water quality impairment. IBG-DCE is computed as:  $\text{TR} + \text{IG} - 1$  (with IBG-DCE  $> 20$  set to 20). IBG-DCE values varies between 1 (low quality) and 20 (high quality). The categories of ecological status for the IBG-DCE are bad (note of 0–5), poor (5–9), fair (9–12), good (12–14) and high (14–20). The I2M2 is made up of 3 elementary metrics considering the sensitivity to anthropogenic disturbance: Average Score Per Taxon or ASPT, relative abundance of ovoviparous and multivoltine taxa in the assemblage; and 2 metrics describing the taxonomic assemblage structure: Shannon diversity and taxonomic richness. The I2M2 is compatible with the requirements of the Water Framework Directive, as it considers the stream reach typology, includes the calculation of a deviation from a reference state and is expressed in Ecological Quality Ratio from 1 (assemblage conforming to the reference) to 0 (completely disturbed). According to the classification rules for the Albarine, the categories of ecological status for the I2M2 values are bad ( $< 0.118$ ), poor (0.118–0.236), fair (0.236–0.354), good (0.354–0.605) and high ( $> 0.605$ ). It should be finally noted that IBG-DCE invertebrates uses the family level as the finest level of identification whereas for I2M2 it is the genus.

#### 2.5.3. Biological traits

The biological traits were extracted from the European freshwater invertebrate database (Tachet et al., 2010). We considered 10 traits: maximal size, life span, number of generations per year, types of reproduction, forms of resistance, types of respiration, dispersal, types of locomotion, food and feeding habits. In this database, traits are divided into categories and are quantified using fuzzy coding (Chevene et al., 1994), which consists in giving an affinity score to a given trait category, between 0 and 3 (maximal size, life span, number of generations per year, types of reproduction, forms of resistance, types of respiration, dispersal, types of locomotion) or 0 and 5 (food and feeding habits), for each taxon. Fuzzy coding enables the compilation of information from various sources and accounts for within-taxon variations.

## 2.6. Data analysis

To quantify the effects of the sewage treatment plants, we compared the concentrations in nutrients and associated parameters between downstream and upstream effluent releases. For each physical and chemical parameter, we calculated the difference between the downstream and upstream values at each reach and each season. Positive values indicated a higher degradation caused by sewage effluents. For each physical and chemical parameter, we tested if the mean difference was significantly greater than zero using unilateral paired t-tests. To assess the efficiency of the overall family richness, IBG-DCE and I2M2 at detecting water quality degradation, we used a mixed-effect model with sewage (up or downstream effluent releases), season (spring vs. fall) and their interaction as fixed effects, and reaches as random effects. The significance of mixed-effect models was evaluated by a comparison with its associated null model using likelihood-ratio tests. All mixed-effect models were computed with the function 'lmer' of package lme4 (Bates et al., 2019). We checked whether the random effects were significant using the function 'rand' of package lmerTest (Kuznetsova et al., 2020). We performed four separate analyses considering overall, EPT, OCH, and Gastropods, Oligochaeta, and Diptera (GOLD) richness. These groups were chosen because they are known to respond differently to FI (e.g. Bonada et al., 2006b; Datry et al., 2016).

We tested our first hypothesis on the negative influence of FI on family richness, IBG-DCE and I2M2 by performing a multiple linear regression with FI and season (spring vs. fall) as explanatory variables.

To test our second hypothesis that the severity of flow intermittence would impact the efficiency of pollution detection, we ran the same tests only on sites upstream effluent releases. We also computed the differences (hereafter delta) in family richness, IBG-DCE and I2M2 between up and downstream effluent releases at each reach and each sampling date. If sewage effects were detected, we would expect positive delta values for OCH and GOLD taxa richness, but negative delta values for the overall family richness, EPT richness, and for the IBG-DCE and I2M2 metrics. We assessed the effect of FI on these delta values with a multiple linear regression, using FI and season (spring vs. fall) as explanatory variables.

To further analyze how intermittence influenced our ability to detect the effects of sewage on biota, we calculated the absolute difference in abundance between up and downstream effluent releases for each taxon. These delta-abundances were log-transformed to reduce the disparities between taxa. We performed a Principal Component Analysis (PCA) on the log-delta abundance values using the 'dudi.pca' function of package ade4 (Dray et al., 2007; Thioulouse et al., 2018). The effect of flow regime (FR; perennial vs. intermittent) and season on the taxon delta-abundances was assessed by running Monte-Carlo tests on the proportion of variance explained between groups (with groups determined either by FR or by season; Dolédec and Chessel, 1987) using the functions 'bca' and 'randtest' of package ade4. These patterns were further tested by a two-way ANOVA on the mean log-delta abundance values of all taxa for each reach and each season, with FR, season and their interaction as explanatory variables.

Finally, to test our third hypothesis on the influence of sewage and FI on the trait combination in communities, we calculated the Community Weighted Mean (CWM) array using the 'functcomp' function of package FD (Laliberté et al., 2014). Then, we computed a two-way ANOVA on the CWM values of each trait category, using sewage (up and downstream effluent releases), FR (intermittent vs. perennial) and their interaction as explanatory variables.

All analyses were performed using the R software (R Core Team, 2021).

## 3. Results

### 3.1. Flow and physical and chemical characteristics of the studied reaches

Our dataset comprised two perennial reaches and four intermittent reaches, the latter drying from 30% to 60% of the time within a year (Table 1). The difference between up and downstream effluent releases values for all physical and chemical parameters were strictly positive in 64% of all reaches and seasons. The average difference was positive for total organic carbon, dissolved organic carbon, total phosphorus and orthophosphate (Appendix A). This indicates that the sewage treatment plants altered the water quality of the stream. In contrast, the average differences between up and downstream effluent releases values were not influenced by the flow regime (Appendix B).

### 3.2. Biological indices

A weak difference between up and downstream effluent releases values occurred only for the GOLD richness (about 1 taxa) (Table 2). In contrast, all biological metrics decreased strongly with increasing FI except the OCH and GOLD taxa richness (Table 3, Fig. 3). This holds when considering samples made only upstream effluent releases for GOLD taxa richness, which was also influenced by season (Tables 2, 3). Except for OCH taxa richness, the delta values between up and downstream effluent releases did not respond to the FI gradient (Table 3). We found a higher number of OCH taxa down than upstream effluent releases for the more intermittent reaches. Last, all variables were lower in fall compared to spring (Table 2).

A two-way ANOVA with FR (intermittent or perennial) and season (spring or fall) as factors indicated that two delta values responded to season (Table 4, Fig. 4). For instance, GOLD taxa were more numerous up than downstream effluent releases during spring (delta value:  $-1.20 \pm 2.17$ ) and it was the opposite during fall ( $1.67 \pm 1.21$ ). The IBG-DCE was slightly higher up than downstream effluent releases during spring ( $0.67 \pm 1.63$ ) and greatly lower during fall ( $-1.80 \pm 1.30$ ). Delta I2M2 values were the only values that responded to FR, with I2M2 being always higher up than downstream effluent releases. Such a difference was higher at perennial ( $-0.160 \pm 0.052$ ) than intermittent reaches ( $-0.033 \pm 0.067$ ). In addition, the I2M2 values classified the intermittent reaches up and downstream effluent releases in "good" ecological status, whereas perennial reaches were classified as "high", upstream effluent releases, and "good", downstream effluent releases.

Four out of the 6 delta values did not respond to FI nor to FR: overall family richness, EPT and GOLD family richness, and IBG-DCE (Table 4, Fig. 4). The overall delta family richness equaled  $\sim$  zero or was even greater than zero (on average  $0 \pm 4$  families), suggesting that family richness was sometimes greater downstream than upstream effluent releases (Table 4, Fig. 4). The delta EPT values were low ( $-1 \pm 1$  taxa), indicating no response of this group to sewage.

### 3.3. Structural responses

The first-two axes of a PCA performed on the delta log-abundances of taxa explained 55.2% of the total variance. The respective between-groups variance equaled 22.4%,  $p = 0.011$  for FR (Fig. 5A), 11.2%,  $p$

**Table 1**

Hydrological characteristics of each sampled reach. FR stands for 'flow regime', FI stands for 'flow intermittence'.

| Reaches           | Distance to the source (km) | FR           | FI (%) |
|-------------------|-----------------------------|--------------|--------|
| Brenod            | 7.2                         | Perennial    | 0      |
| Champdor          | 11.0                        | Intermittent | 60     |
| Hauteville        | 19.3                        | Intermittent | 50     |
| Chaley            | 23.7                        | Perennial    | 0      |
| Bettant           | 74.5                        | Intermittent | 30     |
| Amberieu en Bugey | 51.6                        | Intermittent | 40     |

**Table 2**

Results of the mixed-effect models performed on the family richness for all taxa, EPT, OCH and GOLD, and biological indices IBG-DCE and I2M2, as response variables, sewage (up vs. downstream effluent releases), season (spring vs fall) and their interaction as fixed effects, and reaches as random effects. The random effect was significant for all variables except GOLD richness. EPT stands for Ephemeroptera, Plecoptera, Trichoptera; OCH for Odonata, Coleoptera, Heteroptera; GOLD for Gastropods, Oligochaeta and Diptera. 'df' stands for degrees of freedom.

| Variable        | Degradation   |                 |    |      |         | Season       |              |    |       |           | Interaction |      |          |
|-----------------|---------------|-----------------|----|------|---------|--------------|--------------|----|-------|-----------|-------------|------|----------|
|                 | Mean upstream | Mean downstream | df | Chi2 | p-value | Mean spring  | Mean fall    | df | Chi2  | p-value   | df          | Chi2 | p-value  |
| Family richness | 25(±8)        | 27(±7)          | 2  | 0.61 | 0.738   | 30(±6)       | 22(±7)       | 2  | 10.12 | 0.006 **  | 1           | 0.04 | 0.838    |
| EPT richness    | 6(±2)         | 5(±3)           | 2  | 4.57 | 0.102   | 7(±2)        | 4(±2)        | 2  | 15.39 | 0.000 *** | 1           | 0.63 | 0.427    |
| OGH richness    | 4(±2)         | 5(±3)           | 2  | 1.88 | 0.391   | 5(±3)        | 3(±2)        | 2  | 7.224 | 0.027 *   | 1           | 0.03 | 0.861    |
| GOLD richness   | 9(±1)         | 9(±2)           | 2  | 8.21 | 0.016 * | 9(±1)        | 9(±2)        | 2  | 7.751 | 0.021 *   | 1           | 7.22 | 0.007 ** |
| IBG-DCE         | 13.3(±3.4)    | 13.5(±3.3)      | 2  | 1.22 | 0.544   | 14.7(±2.2)   | 12.0(±3.8)   | 2  | 10.39 | 0.006 **  | 1           | 1.19 | 0.276    |
| I2M2            | 0.50(±0)      | 0.57(±0.15)     | 2  | 5.33 | 0.070   | 0.58 (±0.15) | 0.48 (±0.16) | 2  | 8.851 | 0.012 *   | 1           | 0.04 | 0.838    |

**Table 3**

Results of the linear regression performed on the taxonomic richness for all taxa, EPT, OCH and GOLD, and indicators IBG-DCE and I2M2 as response variables, FI, and season (spring vs. fall) as explanatory variables. All models were computed using all sites, upstream effluent releases only, and on delta values between up and downstream effluent releases. EPT stands for Ephemeroptera, Plecoptera, Trichoptera; OCH for Odonata, Coleoptera, Heteroptera; GOLD for Gastropods, Oligochaeta and Diptera.

|                 | Estimate  | All sites |      |         |            | Reference sites only |      |         |            | Delta impacted-reference |       |         |         |
|-----------------|-----------|-----------|------|---------|------------|----------------------|------|---------|------------|--------------------------|-------|---------|---------|
|                 |           | Estimate  | SD   | t value | p-value    | Estimate             | SD   | t value | p-value    | Estimate                 | SD    | t value | p-value |
| Family richness | Intercept | 35        | 2    | 18.93   | <0.001 *** | 37                   | 2    | 15.75   | <0.001 *** | -2                       | 3     | -0.79   | 0.447   |
|                 | FI        | -19       | 4    | -4.45   | <0.001 *** | -26                  | 5    | -5.02   | 0.001 **   | 14                       | 7     | 2.06    | 0.069   |
|                 | Fall      | -7        | 2    | -3.74   | 0.001 **   | -7                   | 2    | -2.90   | 0.018 *    | -2                       | 3     | -0.52   | 0.614   |
| EPT richness    | Intercept | 9         | 1    | 14.33   | <0.001 *** | 9                    | 1    | 11.03   | <0.001 *** | -1                       | 1     | -0.94   | 0.371   |
|                 | FI        | -7        | 1    | -4.70   | <0.001 *** | -7                   | 2    | -3.51   | 0.007 **   | 0                        | 2     | 0.00    | 1.000   |
|                 | Fall      | -2        | 1    | -3.74   | 0.001 **   | -2                   | 1    | -2.32   | 0.046 *    | -1                       | 1     | -1.36   | 0.208   |
| OGH richness    | Intercept | 6         | 1    | 6.98    | <0.001 *** | 6                    | 1    | 9.74    | <0.001 *** | -1                       | 1     | -0.67   | 0.521   |
|                 | FI        | -2        | 2    | -1.27   | 0.219      | -5                   | 1    | -3.66   | 0.005 **   | 6                        | 2     | 2.42    | 0.039 * |
|                 | Fall      | -2        | 1    | -2.13   | 0.045 *    | -2                   | 1    | -2.74   | 0.023 *    | 0                        | 1     | -0.31   | 0.763   |
| GOLD richness   | Intercept | 9         | 1    | 14.67   | <0.001 *** | 9                    | 1    | 12.78   | <0.001 *** | -1                       | 1     | -0.50   | 0.626   |
|                 | FI        | -1        | 1    | -0.97   | 0.345      | -1                   | 2    | -0.57   | 0.581      | -1                       | 2     | -0.48   | 0.640   |
|                 | Fall      | 0         | 1    | 0.62    | 0.541      | -1                   | 1    | -1.32   | 0.219      | 3                        | 1     | 2.40    | 0.040 * |
| IBGN            | Intercept | 17.7      | 0.7  | 24.15   | <0.001 *** | 17.5                 | 1.3  | 13.58   | <0.001 *** | 0.62                     | 0.88  | 0.70    | 0.500   |
|                 | FI        | -10.1     | 1.7  | -6.09   | <0.001 *** | -10.5                | 2.9  | -3.61   | 0.006 **   | 0.16                     | 1.99  | 0.08    | 0.939   |
|                 | Fall      | -2.8      | 0.8  | -3.54   | 0.002 **   | -2.0                 | 1.3  | -1.49   | 0.170      | -2.17                    | 0.92  | -2.36   | 0.043   |
| I2M2            | Intercept | 0.72      | 0.04 | 17.36   | <0.001 *** | 0.79                 | 0.05 | 14.61   | <0.001 *** | -0.116                   | 0.054 | -2.13   | 0.062   |
|                 | FI        | -0.46     | 0.09 | -4.91   | <0.001 *** | -0.55                | 0.12 | -4.56   | 0.001 **   | 0.147                    | 0.122 | 1.20    | 0.261   |
|                 | Fall      | -0.10     | 0.04 | -2.35   | 0.029 *    | -0.10                | 0.06 | -1.78   | 0.108      | -0.045                   | 0.057 | -0.80   | 0.446   |

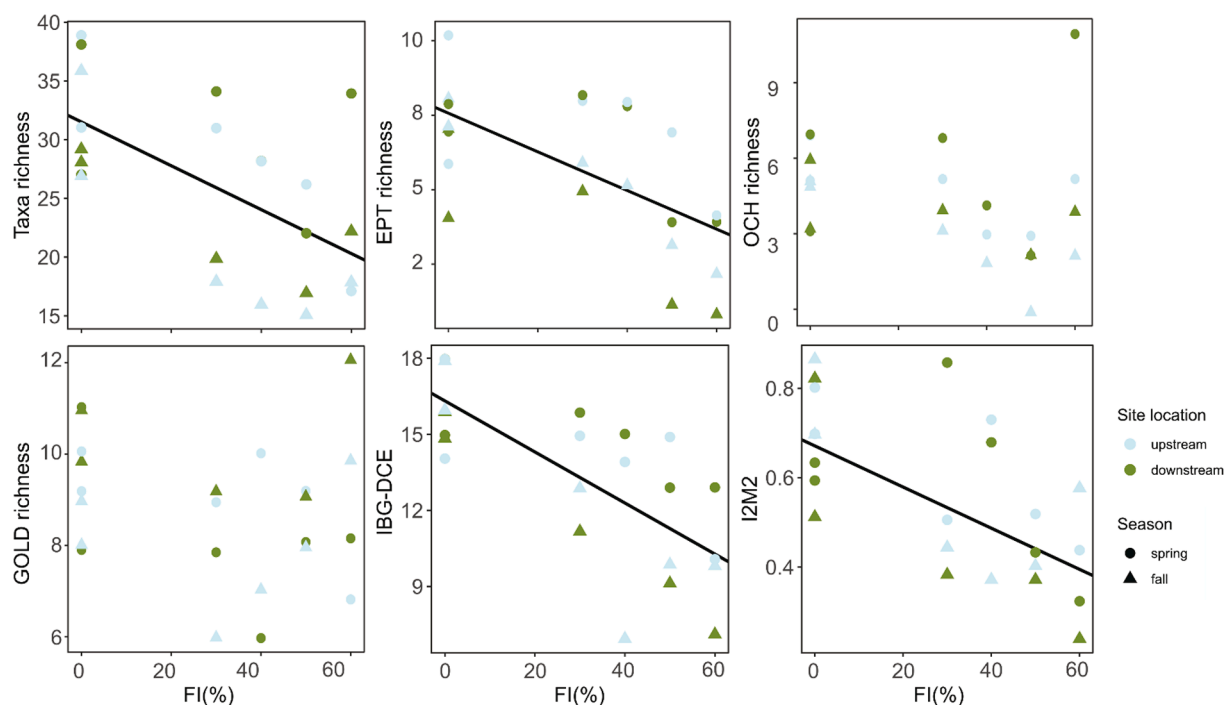
= 0.281 for the season (Fig. 5B) and 40.5%,  $p = 0.049$  for their combination (Fig. 5C). The difference in abundances between up and downstream effluent releases were lower intermittent reaches compared to perennial reaches ( $0.57 \pm 0.24$  and  $1.03 \pm 0.22$  respectively, Table 5). These differences were also lower in fall compared to spring ( $0.58 \pm 0.34$  and  $0.87 \pm 0.26$  respectively, Table 5).

### 3.4. Functional responses

There was no difference in affinity for any trait category between up and downstream effluent releases (Appendix C), whereas affinity differed between intermittent and perennial reaches for 11 trait categories belonging to six different traits. Perennial reaches were characterized by higher proportions of organisms with long life span, using ovoviviparity and passive aquatic dispersal. They were also characterized by lower proportions of organisms that were univoltine, with aerial active dispersal, resistance forms, using cemented clutches and feeding on living macrophytes compared to intermittent reaches (Appendix C). The affinity of only one trait category differed between up and downstream effluent releases according to the FI: at intermittent reaches, the proportion of organisms consuming dead plants (>1mm) was higher upstream than downstream effluent releases at intermittent reaches and the opposite was true at perennial reaches.

## 4. Discussion

In this study, we quantified the structural and functional responses to point-source sewage effluents along a significant gradient of flow intermittence (FI; from 0 to 60%). As expected, the effect of drying confounded with that of water quality degradation and in most cases, current biomonitoring metrics did not distinguish between the effect of sewage and those linked to drying. As all community metrics decreased linearly with increased FI, our second prediction was partly validated and current biomonitoring approaches could be appropriately applied to reaches prone to weak flow intermittence. Last, we identified six biological traits responding to FI but not to water quality degradation. Confirming previous suggestions (Morais et al., 2004; Soria et al., 2020; Stubbington et al., 2018), these results validate that current biomonitoring approaches are unsuitable for application in intermittent rivers. If intermittent rivers with very short (i.e. < 30 days per year) periods of drying could still be monitored as their perennial counterparts, alternative biomonitoring approaches are needed for intermittent rivers with longer periods of drying (i.e. > 30 days per year), along with additional work to define a safe operational space for the application of a standardized biomonitoring approaches in intermittent rivers (Gallart et al., 2017; Munné et al., 2021).



**Fig. 3.** Macroinvertebrate family richness for the overall, EPT, OCH and GOLD taxa, and for biological indices (IBG-DCE and I2M2) along a gradient of flow intermittence (noted FI). The lines represent the significant fits of the mixed models. Circles and triangles represent spring and fall samples respectively, and blue and green colors stand for sites upstream and downstream releases respectively. EPT stands for Ephemeroptera, Plecoptera, Trichoptera; OCH for Odonata, Coleoptera, Heteroptera; GOLD for Gastropods, Oligochaeta and Diptera.

**Table 4**

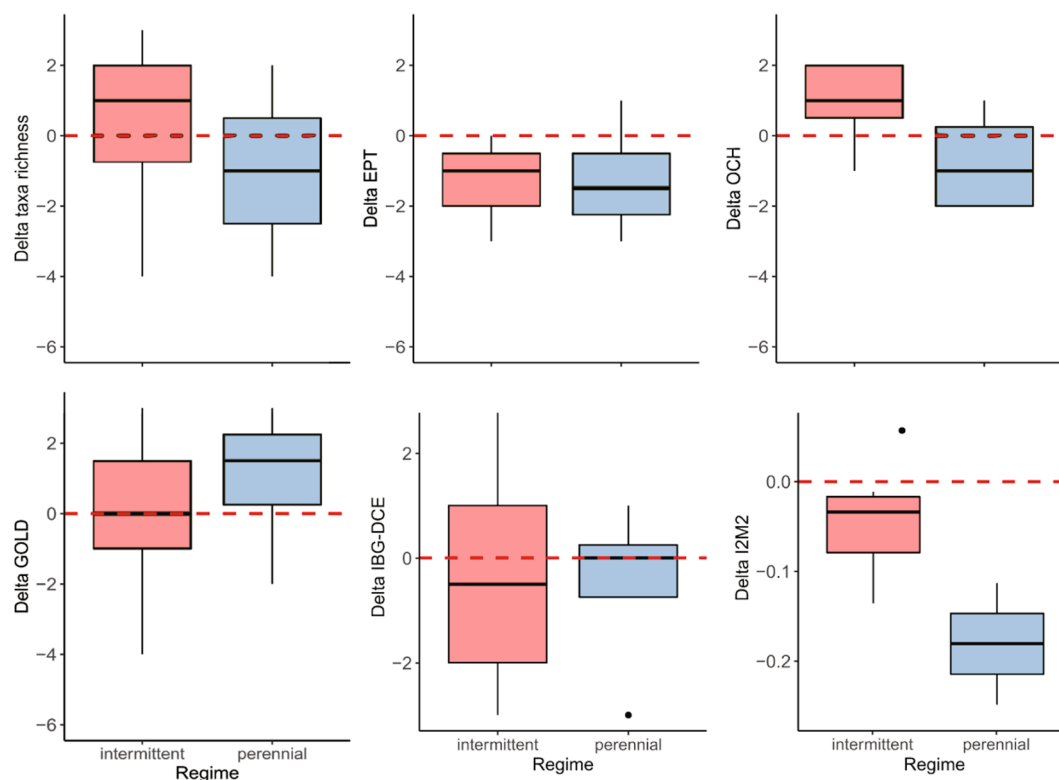
Two-way ANOVA results showing the effects of season, flow regime (FR) and their interaction on the delta values (difference between up and downstream effluent releases) of the overall, EPT, OCH and GOLD family richness, and biological indices (IBG-DCE and I2M2). EPT stands for Ephemeroptera, Plecoptera, Trichoptera; OCH for Odonata, Coleoptera, Heteroptera; GOLD for Gastropods, Oligochaeta and Diptera.

| Delta value   | Effect    | Df | Mean Sq | F-value | P     |
|---------------|-----------|----|---------|---------|-------|
| Taxa richness | FR        | 1  | 33      | 2.52    | 0.156 |
|               | Season    | 1  | 3       | 0.27    | 0.623 |
|               | FR:Season | 1  | 4       | 0.32    | 0.589 |
|               | Residuals | 7  | 13      |         |       |
| EPT richness  | FR        | 1  | 0       | 0.00    | 0.968 |
|               | Season    | 1  | 2       | 1.06    | 0.337 |
|               | FR:Season | 1  | 1       | 0.33    | 0.585 |
|               | Residuals | 7  | 2       |         |       |
| OCH richness  | FR        | 1  | 8       | 3.92    | 0.088 |
|               | Season    | 1  | 1       | 0.42    | 0.590 |
|               | FR:Season | 1  | 0       | 0.00    | 0.964 |
|               | Residuals | 7  | 2       |         |       |
| GOLD richness | FR        | 1  | 3       | 0.85    | 0.388 |
|               | Season    | 1  | 24      | 7.86    | 0.026 |
|               | FR:Season | 1  | 1       | 0.47    | 0.514 |
|               | Residuals | 7  | 21      |         |       |
| IBGN          | FR        | 1  | 0.0     | 0.01    | 0.948 |
|               | Season    | 1  | 16.6    | 5.89    | 0.046 |
|               | FR:Season | 1  | 0.4     | 0.13    | 0.733 |
|               | Residuals | 7  | 2.8     |         |       |
| I2M2          | FR        | 1  | 0.04    | 9.25    | 0.019 |
|               | Season    | 1  | 0.00    | 0.04    | 0.851 |
|               | FR:Season | 1  | 0.00    | 0.82    | 0.396 |
|               | Residuals | 7  | 0.00    |         |       |

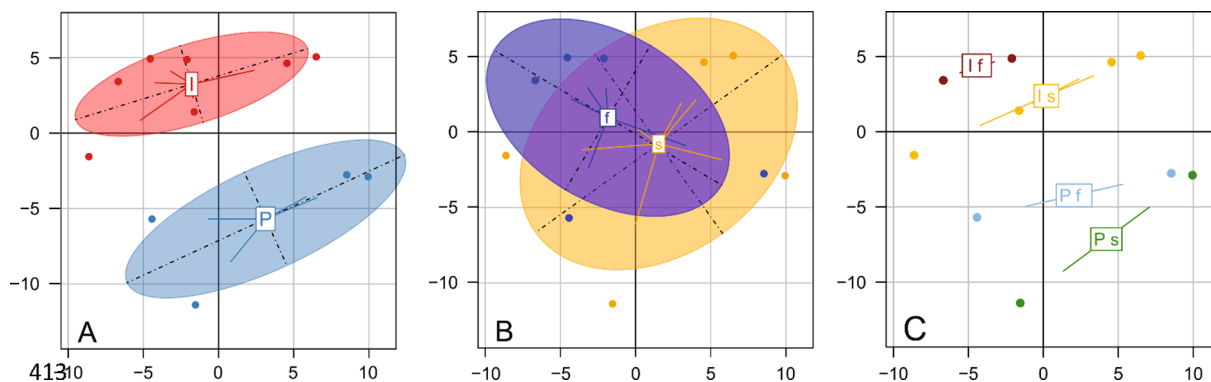
#### 4.1. Current macroinvertebrate-based biomonitoring tools are not suitable in intermittent rivers

Our overarching hypothesis that the ecological status inferred by macroinvertebrate-based biomonitoring protocols is poorly related to water quality of naturally intermittent rivers was largely supported through several lines of circumstantial evidence. First, taxonomic richness was not different up and downstream effluent releases, and the difference in abundances between up and downstream effluent releases were lower at intermittent reaches compared to perennial ones. Consequently, and not surprisingly, IBG-DCE did not differ between up and downstream effluent releases at intermittent reaches. The I2M2 scores were slightly lower downstream effluent releases, but this was largely due to the differences detected in perennial reaches, and not intermittent ones.

We found no differences between up and downstream effluent releases in terms of the relative presence of any trait category tested here. The two trait categories considered in the I2M2, namely multivoltinism and ovoviviparity, were selected for this biomonitoring tool because they allowed a good trade-off between the discrimination efficiency, the specificity and the stability of the metric, and because these trait categories were known to respond to changes in water quality or habitat degradation (Mondy et al., 2012). Indeed, a high number of reproductive cycles is a resilience strategy in ecosystems prone to disturbance (Townsend and Hildrew, 1994). Ovoviviparity might prevent high egg mortality rate as they are protected from harsh conditions (e.g. Dolédec and Statzner, 2008; Piscart et al., 2006), and it has been shown to occur prominently during dry seasons (Beche et al., 2006). However, for these two trait categories we found no difference in their proportions between up and downstream effluent releases. Voltinism varied between perennial and intermittent reaches but we did not observe higher multivoltinism at intermittent reaches, although it is sometimes considered as a strategy for aquatic macroinvertebrates to persist in the context of flow intermittence (Bonada et al., 2007; Stubbington et al., 2017). The negative relationships between ovoviviparity and FI is likely due to



**Fig. 4.** Delta (difference between up and downstream effluent releases) for the overall, EPT, OCH and GOLD family richness and biological indicators (IBG-DCE and I2M2) in intermittent (red) and perennial (blue) reaches. The red dashed line stands for no effect of sewage. EPT stands for Ephemeroptera, Plecoptera, Trichoptera; OCH for Odonata, Coleoptera, Heteroptera; GOLD for Gastropods, Oligochaeta and Diptera.



**Fig. 5.** Projection of the samples upon the first two axes of a PCA performed on the log-delta abundances (absolute difference between up and downstream effluent releases) grouped by FR (A), season (B), and both (C). I and P stands for “intermittent” and “perennial” respectively, f and s stands for “fall” and “spring” respectively.

**Table 5**

Results of a two-way ANOVA performed on the overall log-delta abundances (difference between up and downstream effluent releases).

|           | Df | Sum Sq | Mean Sq | F     | P     |    |
|-----------|----|--------|---------|-------|-------|----|
| FR        | 1  | 0.56   | 0.56    | 22.34 | 0.002 | ** |
| Season    | 1  | 0.28   | 0.28    | 11.11 | 0.013 | *  |
| FR:Season | 1  | 0.03   | 0.03    | 1.02  | 0.347 |    |
| Residuals | 7  | 0.18   | 0.03    |       |       |    |

ovoviviparous (or partially ovoviviparous) taxa collected that are highly dependent on flow permanence (e.g. *Gammarus* and *Pisidium*). Hence, using ovoviviparity in biomonitoring methods in intermittent rivers might lead to unexpected patterns. In contrast, water quality degradation and drying had combined effects on the proportion of organisms

that feed on dead plants, which were scarce (8 taxa per sample) up and downstream effluent releases of intermittent reaches, but were found in higher proportions upstream effluent releases at perennial reaches (14 taxa per sample). The lack of trait responses to water quality degradation suggests that even when accounting for biological traits, current biomonitoring metrics (i.e. I2M2) are poorly discriminant in intermittent rivers, partly due to a certain degree in the functional redundancy observed in aquatic macroinvertebrate communities from naturally intermittent rivers due to a strong environmental filtering (Crabot et al., 2021a). While previous work highlighted that the I2M2 was more responsive to chemical water conditions compared to hydro-morphological alterations (Corneil et al., 2018), our study suggests that drying is an extreme case of flow regime alteration which can hinder the detection of the effect of sewage.

Previous studies have suggested that applying current biomonitoring

approaches in a context of flow intermittence can be challenging (e.g. Cid et al., 2020; Datry et al., 2016; Morais et al., 2004; Reyjol et al., 2014; Steward et al., 2012; Stubbington et al., 2018). To our knowledge, this is the first time that a dedicated study confirms this challenging issue using macroinvertebrate-based metrics along a gradient of flow intermittence. More than half of the worldwide river network is prone to flow intermittence (Messager et al., 2021), being as high as 35% in France (Beaufort et al., 2018; Snelder et al., 2013), with obviously profound implications in terms of management in the context of the Water Framework Directive in Europe. Many water bodies can be considered as in poor or mediocre state while they are in fact in a good ecological status. For example, in our study, the abundance of indicator species including species from several EPT families such as Odontoceridae (Trichoptera), Chloroperlidae and Capniidae (Plecoptera) decreased with pollution in perennial reaches but not in intermittent reaches.

#### 4.2. Defining an operational spaces for assessing the ecological status of intermittent rivers

Our second prediction was only partly supported, which suggests that identifying thresholds in FI for shifting from current macroinvertebrate-based biomonitoring approaches to some alternative is not straightforward. Confirming previous studies carried out in other intermittent rivers (e.g. Crabot et al., 2021a; Datry et al., 2016; Piano et al., 2020), we found a decrease in taxonomic richness, the number of EPT taxa, and the two French biomonitoring indices with increasing FI. However, such relationships were significant only upstream effluent releases; we found a linear effect as shown by others (e.g. Datry et al., 2016) but our limited number of reaches ( $n = 6$ ) may hinder the identification of tipping-points in those relationships. This indicates thresholds in FI may be difficult to identify and suggests that defining safe operational spaces for applying current macroinvertebrate-based biomonitoring approaches in intermittent rivers may be a difficult task. By the way, downstream effluent releases, the combined effects of sewage and FI hindered these relationships.

We also found a strong temporal variability in community responses to FI and sewage, corroborating the view of intermittent rivers as being highly dynamic (Datry et al., 2016; Sarremejane et al., 2017a), which makes the assessment of the ecological status of these prevalent ecosystems even more complex. The delta values in taxonomic richness, number of OCH taxa and the IBG-DCE were higher in fall, after the dry season compared to the spring pre-drying season, in accordance with previous results showing difference in biomonitoring scores between seasons on intermittent streams in reference conditions (Munné and Prat, 2011). This is most likely due to legacy effects of drying events on aquatic communities and processes, as recovery upon rewetting can take several months when observed (e.g., Datry, 2012; Piano et al., 2020). However, even in spring, before the dry season, the persisting effect of drying was still too strong to allow detection of the effects of sewage effluents. Nevertheless, in some cases, the organic pollution was dramatic, with as high as 9 mg/L of DOC and up to 31,133 MPN/100 mL for *E. coli* (respectively twice and 26 times higher than the values found upstream effluent releases of the same reach). But even in this case, the responses of community and biomonitoring metrics were not different downstream compared to upstream the effluent release due to the drying regime of the reach. These results suggest that when applied before the drying season, current macroinvertebrate-based biomonitoring approaches are poorly applicable to naturally intermittent rivers, which echoes some recent calls to develop alternative approaches (e.g. Cid et al., 2020; Stubbington et al., 2018), either focusing on aquatic biotic groups adapted to desiccation (e.g. using biological traits, Soria et al., 2020), terrestrial and semi-aquatic organisms (e.g. riparian arthropods, Steward et al., 2018) or including regional processes to set up dynamic references (Cid et al., 2020).

For those bioindicators based on a reference condition such as the

I2M2 (Mondy et al., 2012), a first step could be to include new references in the current classification that would be representative of the effect of flow intermittence across ecoregions. Another path would be to better take into account the temporal dynamic of these ecosystems (Cid et al., 2020). Confirming previous observations, our results highlighted the high temporal variation of communities at reaches prone to drying and its influence on biomonitoring methods (Crabot et al., 2020; Leigh et al., 2019; Munné and Prat, 2011). As such, future models built to describe reference conditions at intermittent reaches should include replicates of biotic and abiotic samples at different seasons to capture the temporal variability of community composition and avoid underestimating the diversity at these locations (Cid et al., 2020; Sarremejane et al., 2017a). This could be especially important in Mediterranean headwater streams with regular phases of stagnant disconnected pools, which greatly differ from flowing and dry phases (Bonada et al., 2020; Gallart et al., 2017). This may be challenging because the ideal biomonitoring tools should be designed according to 12 major criteria among with the implementation comprising: low costs for sampling and sorting (field approaches), simple sampling protocol and low cost for taxa identifications (no specialists in taxonomy required) (Bonada et al., 2006a). Therefore, assessing the ecological status of intermittent rivers will surely involve a shift in paradigm.

Many formerly perennial rivers are becoming intermittent (Döll and Schmied, 2012; Messager et al., 2021), raising the question whether current biomonitoring tools would also fail in such novel-ecosystems (*sensu* Hobbs et al., 2013). Whereas we are unaware of empirical data that could help to address this key question, our results suggest that current macroinvertebrate-based approaches would be even less adequate in artificially intermittent rivers, at least during the first times following the change in flow regime. This is because in such contexts, a dramatic decrease in the number of species is expected, along with cascading effects on functional traits due to poor functional redundancy (Aspin et al., 2019; Crabot et al., 2021a). The key question is to determine the trajectory of communities in newly-intermittent rivers, which remain unknown (Crabot et al., 2021b). If community trajectories converge gradually towards those from naturally intermittent rivers, our results indicate substantial improvements will be needed to accurately assess the ecological status of rivers and streams prone to drying. If these trajectories take a different path, we are today unable to foresee which biomonitoring tools will be needed and call for a research agenda taking into account the alterations of flow regimes and appearance of hybrid ecosystems (*sensu* Hobbs et al., 2013) in the Anthropocene era.

#### CRedit authorship contribution statement

**Julie Crabot:** Formal analysis, Writing – original draft. **Sylvain Dolédec:** Formal analysis, Writing – original draft. **Maxence Forcellini:** Formal analysis. **Thibault Datry:** Conceptualization, Methodology, Resources, Writing – original draft.

#### Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Data availability

Data are available on request.

## Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ecolind.2021.108440>.

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